

$$\lambda_{\text{new}} = 2.0 \gamma$$

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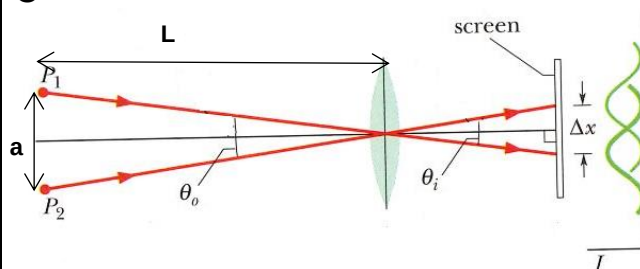
C

$$d \sin \theta = n \lambda, \text{ so } \sin 30^\circ = \lambda/d$$

$$x = \lambda D/a = \lambda D/500d = (1.0/500) \sin 30^\circ = 1.0 \times 10^{-3} \text{ m}$$

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C



P_1 and P_2 are the double stars.

The angular separation θ_o of the 2 stars equals the angular separation θ_i of the images. i.e. $\theta_o = \theta_i$

If the images are to satisfy Rayleigh's criterion for resolvability, then

$$\theta_o = \theta_i = \theta_R = \frac{\lambda}{D}, \text{ where } D = \text{diameter of the lens}$$

Hence $\frac{a}{L} = \frac{\lambda}{D}$, taking $\lambda = 500 \text{ nm}$ for visible light

$$\frac{a}{20 \times 9.5 \times 10^{15}} = \frac{500 \times 10^{-9}}{3.0}$$

$$a = 3.2 \times 10^{10} \text{ m}$$

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